Robust Adaptive Control for a Quadrotor UAV With Uncertain Aerodynamic Parameters

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In controller design for quadrotors, the drag coefficients, thrust coefficients, and aerodynamic damping coefficients are usually unmeasurable, and they have to be regarded as uncertain parameters. Meanwhile, the position information provided by the GPS receiver is easy to get disturbed by environmental noise. A backstepping-based robust adaptive control scheme is proposed in this article to address uncertain parameters, bounded external disturbances, and the noise in measurement. The proposed approach overcomes singularities resulted from using inverse trigonometric functions of Euler angles. A continuous function is introduced to replace signum functions, such that the closed-loop system satisfies Lipschitz condition. Boundedness of estimation errors and tracking errors are guaranteed by the proposed approach and their boundary can be modified. The advantages of the proposed controller are supported by numerical examples and experimental results, where comparisons with proportional-integral-derivative (PID), backstepping, geometric control, and the other existing robust adaptive controller are provided.

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I. INTRODUCTION

Quadrotors are agile flying machines. Due to their agility and full autonomy, they can accomplish many important missions, such as searching and rescuing [1], [2], weather observation [3], aerial delivery [4], [5], firefighting [6], [7], and environment monitoring [8], [9].

A common quadrotor is driven by four motors on it. The kinetic model of a quadrotor is an underactuated nonlinear system with four control inputs and six degrees of freedom. Although the actual inputs are four lifts generated by motors, most of research works choose the sum magnitudes of four lifts and 3-D torques as four inputs for convenience [10], [11], [12]. The transition between two kinds of control inputs is called control allocation, which needs two parameters, namely, drag and thrust coefficients. Both of them are ignored in these research works above. Also, the model with six degrees of freedom can be separated into the position and attitude models. The position model owns three degrees, namely, the position coordinates in x, y, and z-axes. Meanwhile, the attitude model can be written in different ways with three degrees of freedom, including Euler angles, quaternions, and rotation matrix [13]. Therefore, backstepping is suitable for its controller design [10], [12], [14] in order to solve the tracking issues of the underactuated nonlinear system. Furthermore, model uncertainties and external disturbances are inevitable in quadrotor models. Meanwhile, apart from disturbances on the quadrotor, the environmental noises existing in the process of measurement should also be taken into consideration. Numerous related experiments [15], [16] are completed by a highly accurate indoor system-"Optitrack." It uses a lot of cameras to capture the movement of quadrotors and transmits their position signal to them. Although it is quite efficient in experiments, it is not accessible to the majority of individuals due to its high price. Hence, GPS is a more price-friendly way to offer the position information for the masses. The position information of the quadrotor provided by the GPS receiver may get disturbed by the environmental noises [17], [18]. The height information is given by the barometer, which is affected by numerous environmental factors, such as temperature and wind [19], [20]. Hence, the performance is degraded if traditional backstepping is applied. Robust adaptive strategies are applied to designing backstepping controllers for quadrotors to overcome model uncertainties and external disturbances, e.g., [21], [22], [23], [15], etc.

An adaptive integral backstepping control algorithm is proposed to realize robust control of a quadrotoR unmanned aerial vehicle (UAV) with unknown mass [24] and inertia [25]. Another method known as the cerebellar model arithmetic computer (CMAC) provides fast learning and adaptation [23], where a direct approximate-adaptive control is proposed by using CMAC nonlinear approximators. The method updates adaptive parameters and the CMAC weights to achieve both adaptation to unknown payloads and robustness to disturbances. However, beside mass and inertia, there exist other unmeasurable parameters, e.g., the drag coefficients, thrust coefficients, and aerodynamic damping coefficients. Thrust coefficients and drag coefficients are related to the lift and torque generated by motors, and both of them can be calculated indirectly [26]. One difficulty is that only the ranges, instead of accurate values, are provided for the calculation of drag and thrust coefficients in [26], and the calculation is somewhat inaccurate. In another aspect, the accurate calculation of aerodynamic damping coefficient usually requires wind tunnel tests [27], which are usually commercially unavailable.

Some other relevant studies are listed as follows. A robust adaptive control based on backstepping technique is introduced in [15] to deal with the uncertainties of thrust factor and drag factor. The backstepping method in [15] is similar to [28] and [29]. The system is decoupled into two parts: the inner loop for attitude control and the outer loop for position control. In outer loop, a virtual control is designed for the quadrotor UAV to track its reference trajectory; in inner loop, the desired Eular angles are calculated from virtual inputs, and are implemented by thrust and torques. The abovementioned approaches have some shortcomings. First, singularities would exist in inverse trigonometric functions. Second, the assumption that the unknown time-varying disturbances are continuous differentiable is rather strong. Third, the nondifferentiable signum functions in the controller do not satisfy Lipschitz condition and brings challenges to the validity of Lyapunov analysis [30].

In this article, a robust adaptive backstepping controller is introduced to achieve the objective of trajectory tracking. The main contributions are given as follows.

1) Compared to [15], [21], and [22], the uncertain parameters are taken into sufficient consideration, including the moment of inertia, aerodynamic coefficients, drag coefficient, and thrust coefficient. Because of the fact that most of works focus on the thrust and torque rather the actual control inputs, namely, the forces generated by four motors. Hence, to the best of authors' knowledge, the related research works on drag and thrust coefficient in controller design for quadrotors are fairly limited.
2) The controller designed in this article is able to resist the disturbances on the quadrotor. The only requirement for disturbances is that they are bounded, and it is much weaker than existing research works where disturbances are assumed to be continuously differentiable [15] or constants [31]. Furthermore, a series of continuous terms take place of signum functions in [15] so that the system meets Lipshchitz condition and the existence of derivatives of virtual inputs during the design process can be guaranteed.
3) The attitude model is in the form of rotation matrix, which means that it is a kind of geometric model [11], [31]. Different from the model based on Eular angles [12] and [15], the proposed controller does not need the calculation of desired Euler angles, such that singularities resulted from using inverse trigonometric functions can be avoided.
4) Many of current research works on quadrotors fail to consider the effects of environmental noises on the process of measurement, especially the position information of quadrotors because the related experiments are completed indoors [15], [16] by "Optitrack" equipment, which is too expensive for normal research works [5]. However, the position signal of outdoor experiments is provided by a GPS receiver in real time, which is cheaper but easier to get disturbed. In a word, the consideration of noises on the measurement of position information is rather necessary in order to satisfy the needs of most of research works.

Simulation examples are provided to verify the effectiveness of the proposed robust adaptive control scheme. In simulation, the proposed controller are compared with proportional-integral-derivative (PID), backstepping, geometric control, and another robust adaptive method. All of the contributions of the proposed controller can be revealed in the comparison. Also, two different trajectory tracking results of outdoor experiments are provided to testify the feasibility.

The rest of this article is organized as follows. In Section II, the model of a quadrotor UAV is established and the objectives of this article are proposed. In Section III, the robust adaptive controller based on backstepping is introduced and the related proof of stability is given in this section. In Section IV, a set of simulation results of comparison are provided. In Section V, the results of outdoor experiments are offered to test the feasibility of the proposed scheme. Finally, Section VI concludes this article.

II. PROBLEM DESCRIPTION

A. Modeling

The mathematical model of a quadrotor UAV can be divided into two interconnected parts, the position model and the attitude model, and the position model [5], [15], [32], [33] is given by

$$\ddot{x} = -(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)\frac{f}{m} - \frac{K_x}{m}\dot{x} + \frac{d_x}{m} \tag{1}$$

$$\ddot{y} = -(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)\frac{f}{m} - \frac{K_y}{m}\dot{y} + \frac{d_y}{m} \tag{2}$$

$$\ddot{z} = g - (\cos\phi\cos\theta)\frac{f}{m} - \frac{K_z}{m}\dot{z} + \frac{d_z}{m} \tag{3}$$

where $(x, y, z)^{\mathrm{T}}$ stands for the position of the mass center of the UAV in an inertial frame $\{x_I, y_I, z_I\}$, and z_I follows the direction of gravity and points down. ϕ, θ, and ψ are the Euler angles (roll, pitch, and yaw) of the fuselage in body frame $\{x_b, y_b, z_b\}$. The aerodynamic damping coefficients along three axes are defined as K_x, K_y, and K_z. Also, $d_{x,y,z}$ represents the disturbances on the UAV. The magnitude of the combined force produced by four rotors is f, which is also the control input of (1)–(3). m is the mass of the quadrotor UAV, and g stands for the gravity acceleration.

Furthermore, three scalar equations in the position model can be combined into one vector equation as follows:

$$\ddot{X} = \frac{F}{m} - \frac{K_v v}{m} + \frac{d_F}{m} + ge_3$$

where $X = [x, y, z]^{\mathrm{T}}$, $d_F = [d_x, d_y, d_z]^{\mathrm{T}}$, $K_v = \mathrm{diag}(K_x, K_y, K_z)$, and $e_3 = [0, 0, 1]^{\mathrm{T}}$. Meanwhile, F is the combined force working on the quadrotor, and its magnitude is f. v is the velocity vector of the quadrotor. However, v is not simply a derivative of X in fact because of the disturbances on sensors during the measurement of X. For example, x and y are provided by the GPS receiver, which gets disturbed by the environmental noises [17], [18]. z is given by the barometer, which is affected by numerous environmental factors, such as temperature and wind [19], [20]. Therefore, the complete position model of a quadrotor can be written as follows:

$$\begin{aligned}\dot{X} &= v + d_X \\ \dot{v} &= \frac{F}{m} - \frac{K_v v}{m} + \frac{d_F}{m} + ge_3\end{aligned} \tag{4}$$

where d_F stands for the disturbance of the measurement of X.

In addition, the attitude model can be obtained in the form of rotation matrix as follows according to [11]:

$$\begin{aligned}\dot{R} &= R\mathrm{Skew}\,(\Omega) \\ J\dot{\Omega} &= -\Omega \times J\Omega + M + d_M.\end{aligned} \tag{5}$$

$\mathrm{Skew}(\cdot)$ is the skew-symmetric matrix

$$\mathrm{Skew}(w) = \begin{bmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{bmatrix}$$

such that $w \times q = \mathrm{Skew}(w)q$ for any two vectors $w = [w_1, w_2, w_3]^{\mathrm{T}}$ and $q = [q_1, q_2, q_3]^{\mathrm{T}}$. Meanwhile, J is the inertia matrix of the quadrotor, R stands for the rotation matrix, and Ω represents the angular velocity vector. The rotational matrix belongs to special orthogonal group, namely, $R \in SO(3)$ and $R^{\mathrm{T}}R = I$ [11]. Also, the attitude model is influenced by the disturbance d_M. The torque generated by the quadrotor is defined as M, which is the control input of (5). Nevertheless, there are several differences between the situation in the experiment and (5). The actual control inputs in the experiment are the magnitude of forces generated by four rotors, which are defined as F_{1-4}. The relationship between M, f, and F_{1-4} is different in terms of the configuration of the quadrotor. There are mainly two configurations for the quadrotor UAVs, namely, (i) "+" and (ii) "X." The quadrotor mentioned in this article is of the "X" configuration. Thus, the expressions of f and M are

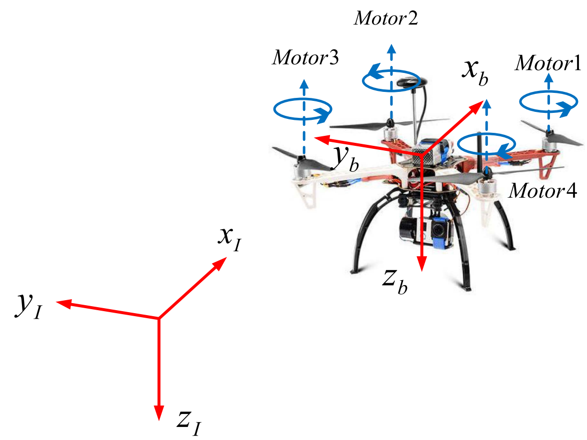

Fig. 1. Model of a quadrotor UAV. The inertial frame and the distribution of motors are labeled in the figure.

defined according to [13] as follows:

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{\sqrt{2}l}{2} & \frac{\sqrt{2}l}{2} & \frac{\sqrt{2}l}{2} & -\frac{\sqrt{2}l}{2} \\ \frac{\sqrt{2}l}{2} & -\frac{\sqrt{2}l}{2} & \frac{\sqrt{2}l}{2} & -\frac{\sqrt{2}l}{2} \\ \frac{c_M}{c_T} & \frac{c_M}{c_T} & -\frac{c_M}{c_T} & -\frac{c_M}{c_T} \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix} \tag{6}$$

where $M = [M_x, M_y, M_z]^{\mathrm{T}}$, l is the distance between the geometric center of the quadrotor and one of four rotors. Meanwhile, c_T and c_M are thrust and drag coefficients, respectively. Hence, the attitude model can be written as follows:

$$\begin{aligned}\dot{R} &= R\mathrm{Skew}\,(\Omega) \\ J\dot{\Omega} &= -\Omega \times J\Omega + \mathrm{diag}\left(l, l, \frac{1}{c}\right) M_1 + d_M\end{aligned} \tag{7}$$

where for any vector $q = [q_1, q_2, q_3]^{\mathrm{T}}$, $\mathrm{diag}(q)$ stands for the following statement:

$$\mathrm{diag}(q) = \begin{bmatrix} q_1 & 0 & 0 \\ 0 & q_2 & 0 \\ 0 & 0 & q_3 \end{bmatrix}.$$

c is chosen as $c = \frac{c_T}{c_M}$ and M_1 is the actual control input of (7) which is defined as

$$M_1 = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}. \tag{8}$$

Also, f can be written as follows:

$$f = F_1 + F_2 + F_3 + F_4 \tag{9}$$

where F_i is the magnitude of the lift generated by the ith motor and $F_i \geq 0$.

B. Objective

There are two main issues to be addressed in this article. First of all, the thrust coefficient c_T, the drag coefficient c_M,

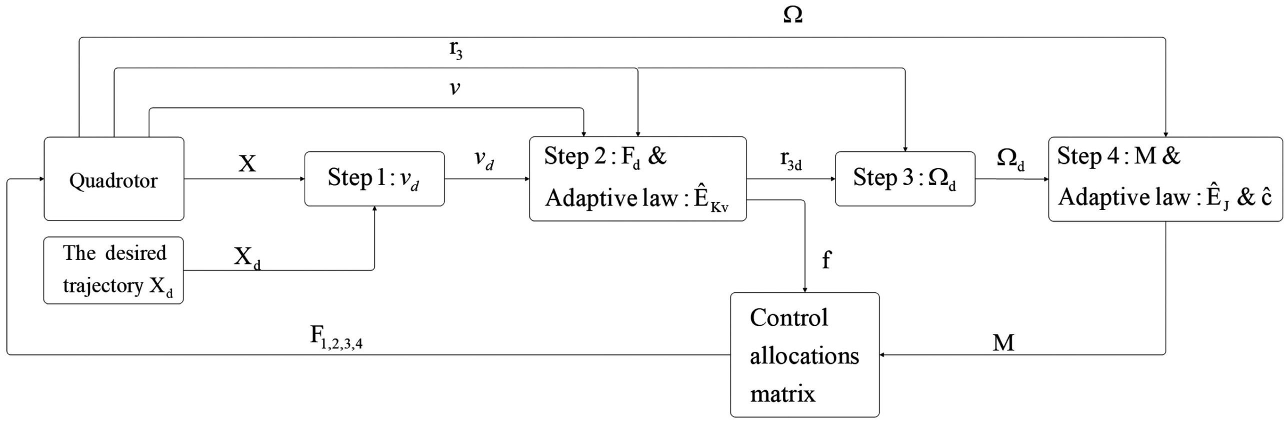

Fig. 2. Structure of the overall control system.

the aerodynamic damping coefficient K_v, and the inertia matrix J cannot be accurately measured according to Section I. Controllers of UAVs are usually in the form of "PID" in practice, where uncertainties in c_T and c_M can be compensated for by adjusting "PID" parameters [26]. The tedious process of tuning PID parameters can be avoided and the issue can be solved at a theoretical level by introducing the adaptive control in this article. Second, the unknown disturbances d_X, d_F, and d_M exist in the kinetic models (4) and (7). Thus, the controller designed in this article should be able to resist disturbances. Moreover, the disturbances are time-varying but bounded.

The *objective* of this article is to design the robust adaptive controller, such that the position of the UAV X can track its desired trajectory X_d in the presence of parameter uncertainties and external disturbances.

III. CONTROLLER DESIGN

Parameters c, J, and K_v are unable to be obtained accurately. Therefore, adaptive control is necessary to solve this problem. Meanwhile, the kinetic model is an underactuated system [34] with four inputs and six degrees of freedom. A feasible controlled output can be X [10]. The control law is designed within backstepping framework and in the form geometric controller, such as [11] and [31]. The block diagram of overall control system is shown in Fig. 2.

ASSUMPTION 1 The reference trajectory X_d is bounded. The third-order derivatives of X_d is bounded and piecewise continuous [11], [10].

ASSUMPTION 2 The disturbances d_X, d_F, and d_M are time-varying and bounded. Their bounds are defined as follows: $\|d_X\|_2 \leq d_{X\max}$, $\|d_F\|_2 \leq d_{F\max}$, and $\|d_M\|_2 \leq d_{M\max}$ [31].

REMARK 1 Compared with [15], the assumption that disturbances d_i are continuously differentiable is no longer needed. The only requirement for $d_{X,F,M}$ is that it is bounded.

A. Adaptive Backstepping Design

The proposed controller is design in the framework of backstepping, where adaptive laws are applied to estimate uncertain inertial and aerodynamic parameters.

Step 1: Define the position tracking error by

$$z_p = X - X_d.$$

The virtual system for the first step is

$$\dot{z}_p = v_d + d_X - \dot{X}_d$$

where v_d is the virtual control input. Let the Lyapunov function at this stage $V_1 = h_p \frac{1}{2} z_p^{\mathrm{T}} z_p$, where h_p is a positive constant and its time derivative is

$$\begin{aligned}\dot{V}_1 &= h_p z_p^{\mathrm{T}} d_X + h_p z_p^{\mathrm{T}} \left(v_d - \dot{X}_d\right) \\ &\leq h_p \|z_P\|_2^2 \|d_X\|_2^2 + h_p z_p^{\mathrm{T}} \left(v_d - \dot{X}_d\right) \\ &\leq \frac{h_p \|z_P\|_2^2 \|d_X\|_2^2}{2} + \frac{h_p}{2} + h_p z_p^{\mathrm{T}} \left(v_d - \dot{X}_d\right) \\ &\leq \frac{h_p \|z_P\|_2^2 d_{X\max}^2}{2} + \frac{h_p}{2} + h_p z_p^{\mathrm{T}} \left(v_d - \dot{X}_d\right).\end{aligned}$$

Thus, the virtual input is chosen as

$$v_d = \dot{X}_d - \frac{k_1}{h_p} z_p - \frac{d_{X\max}^2}{2} z_p \tag{10}$$

which makes $\dot{V}_1 \leq -k_1 z_p^{\mathrm{T}} z_p + \frac{h_p}{2}$. k_1 is a positive control gain.

Step 2: The error of velocity $z_v = v - v_d$ is introduced for the second step. The virtual system is

$$\begin{aligned}\dot{z}_p &= v_d + d_X - \dot{X}_d \\ \dot{z}_v &= \frac{F_d}{m} - \frac{K_v v}{m} + \frac{d_F}{m} + g e_3 - \dot{v}_d\end{aligned} \tag{11}$$

where F_d is the virtual control input. There are several terms of disturbances in $\dot{v}_d$. $\dot{v}_d$ can be written as $\dot{v}_d = \ddot{X}_d - (\frac{k_1}{h_p} + \frac{d_{X\max}^2}{2})(v + d_X - \dot{X}_d)$. Furthermore, (11) is given by the following equation:

$$\dot{z}_p = v_d + d_X - \dot{X}_d + z_v$$

$$\dot{z}_v = \frac{F_d}{m} - \frac{K_v v}{m} + \frac{d_F}{m} + ge_3 - \ddot{X}_d + \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right) \cdot \left(v + d_X - \dot{X}_d\right). \tag{12}$$

The unknown parameter K_v is defined as $K_v = \mathrm{diag}(E_{K_v})$, where $E_{K_v} = [K_x, K_y, K_z]^{\mathrm{T}}$ and its estimation is $\hat{E}_{K_v} = [\hat{K}_x, \hat{K}_y, \hat{K}_z]^{\mathrm{T}}$. The error of the estimation is defined as $\tilde{E}_{K_v} = E_{K_v} - \hat{E}_{K_v}$. The Lyapunov function for the virtual system (12) is $V_2 = \frac{h_p}{2} z_p^{\mathrm{T}} z_p + \frac{h_v}{2} z_v^{\mathrm{T}} z_v + \frac{h_{K_v}}{2} \tilde{E}_{K_v}^{\mathrm{T}} \tilde{E}_{K_v}$, where h_v and h_{K_v} are both positive constants. The derivative of V_2 is shown as follows:

$$\begin{aligned}
\dot{V}_2 &= h_p z_p^{\mathrm{T}} \left(v_d + d_X + \dot{X}_d\right) + h_v z_v^{\mathrm{T}} \left(\frac{F_d}{m} - \frac{K_v v}{m} + \frac{d_F}{m}\right.\\
&\quad + ge_3 - \ddot{X}_d + \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right) \left(v + d_X - \dot{X}_d\right)\\
&\quad \left. + \frac{h_p}{h_v} z_p\right) - h_{K_v} \tilde{E}_{K_v}^{\mathrm{T}} \dot{\hat{E}}_{K_v}\\
&= \dot{V}_1 + h_v z_v^{\mathrm{T}} \left(\frac{F_d}{m} - \frac{K_v v}{m} + \frac{d_F}{m} + ge_3 - \ddot{X}_d + \left(\frac{k_1}{h_p}\right.\right.\\
&\quad \left.\left. + \frac{d^2_{X\max}}{2}\right) \left(v + d_X - \dot{X}_d\right) + \frac{h_p}{h_v} z_p\right) - h_{K_v} \tilde{E}_{K_v}^{\mathrm{T}} \dot{\hat{E}}_{K_v}.
\end{aligned} \tag{13}$$

The virtual control inputs and parameter update law are chosen as

$$\begin{aligned}
F_d &= m \left(\frac{\mathrm{diag}(v) \hat{E}_{K_v}}{m} - ge_3 - \frac{h_p}{h_v} z_p + \ddot{X}_d - \left(\frac{k_1}{h_p}\right.\right.\\
&\quad \left. + \frac{d^2_{X\max}}{2}\right) \left(v - \dot{X}_d\right) - \frac{k_2}{h_v} z_v - \frac{1}{2} \left(\frac{d^2_{F\max}}{m} + \left(\frac{k_1}{h_p}\right.\right.\\
&\quad \left.\left.\left. + \frac{d^2_{X\max}}{2}\right) d^2_{X\max}\right) z_v\right)
\end{aligned} \tag{14}$$

$$\dot{\hat{E}}_{K_v} = -\frac{h_v}{h_{K_v} m} \mathrm{diag}(v) z_v \tag{15}$$

where the control gains k_2 and γ_{K_v} are both positive constants. Substitute (14) and (15) into (13) and then $\dot{V}_2$ is written as follows:

$$\begin{aligned}
\dot{V}_2 &= \dot{V}_1 + h_v z_v^{\mathrm{T}} \left(\frac{\mathrm{diag}(v) \hat{E}_{K_v}}{m} - \frac{k_2}{h_v} z_v - \frac{1}{2} \left(\frac{d^2_{F\max}}{m}\right.\right.\\
&\quad \left. + \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right) d^2_{X\max}\right) z_v - \frac{K_v v}{m} + \left(\frac{k_1}{h_p}\right.\\
&\quad \left.\left. + \frac{d^2_{X\max}}{2}\right) d_X + \frac{d_F}{m}\right) + \frac{h_v}{m} \tilde{E}_{K_v}^{\mathrm{T}} \mathrm{diag}(v) z_v\\
&= \dot{V}_1 - k_2 z_v^{\mathrm{T}} z_v + h_v z_v^{\mathrm{T}} \left(\left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right) d_X + \frac{d_F}{m}\right)\\
&\quad - \frac{h_v}{2} \left(\frac{d^2_{F\max}}{m} + \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right) d^2_{X\max}\right) \|z_v\|_2^2\\
&\quad + h_v z_v^{\mathrm{T}} \left(\frac{\hat{K}_v v}{m} - \frac{K_v v}{m}\right) + \frac{h_v}{m} \tilde{E}_{K_v}^{\mathrm{T}} \mathrm{diag}(v) z_v\\
&\le \dot{V}_1 - k_2 z_v^{\mathrm{T}} z_v + h_v \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right) \|d_X\| \|z_v\|\\
&\quad + h_v \frac{\|d_F\| \|z_v\|}{m} - \frac{h_v}{2} \left(\frac{d^2_{F\max}}{m} + \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right)\right.\\
&\quad \left. \cdot d^2_{X\max}\right) \|z_v\|_2^2 - \frac{h_v}{m} z_v^{\mathrm{T}} \tilde{K}_v v + \frac{h_v}{m} z_v^{\mathrm{T}} \mathrm{diag}(v) \tilde{E}_{K_v}\\
&\le \dot{V}_1 - k_2 z_v^{\mathrm{T}} z_v + \frac{h_v}{2m} + \frac{h_v}{2} \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right)\\
&\le -k_1 z_p^{\mathrm{T}} z_p - k_2 z_v^{\mathrm{T}} z_v + \frac{h_p}{2} + \frac{h_v}{2m}\\
&\quad + \frac{h_v}{2} \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right).
\end{aligned} \tag{16}$$

Step 3: The control input of the position model (4) F_d is offered in Steps 1 and 2. Under the influence, X in (4) can track X_d. However, the quadrotor is not fully actuated and its actual control input F has to follow the direction of r_3, where r_3 is the third axis in the body frame and comes from $R = [r_1, r_2, r_3]$. Also, F is the projection of F_d along r_3 [11], [16] and it can be written as $F = r_3 r_3^{\mathrm{T}} F_d$. In addition, the desired signal of r_3 can be given by $r_{3d} = -\frac{F_d}{\|F_d\|}$ and its derivative of time is $\dot{r}_{3d} = -\Pi_{r_{3d}}(\frac{\dot{F}_d}{\|F_d\|})$ [16], where $\Pi_{r_{3d}}(\frac{\dot{F}_d}{\|F_d\|})$ is a projection operator that projects $\frac{\dot{F}_d}{\|F_d\|}$ onto the plane that is orthogonal to r_{3d} with $\Pi_{r_{3d}}(\frac{\dot{F}_d}{\|F_d\|}) = -\mathrm{Skew}^2(r_{3d}) \frac{\dot{F}_d}{\|F_d\|}$. The tracking error of this step is introduced as $z_r = r_3 - r_{3d}$. The virtual system is written as follows:

$$\begin{aligned}
\dot{z}_p &= v_d + d_X - \dot{X}_d + z_v\\
\dot{z}_v &= \frac{F_d}{m} - \frac{K_v v}{m} + \frac{d_F}{m} + ge_3 - \ddot{X}_d + \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right)\\
&\quad \cdot \left(v + d_X - \dot{X}_d\right) + \frac{F - F_d}{m}\\
\dot{z}_r &= \dot{r}_3 - \dot{r}_{3d} = -R\mathrm{Skew}(e_3) \Omega_d - \dot{r}_{3d}
\end{aligned} \tag{17}$$

where Ω_d is the virtual control input, and $F - F_d$ can be written as $F - F_d = r_3 r_3^{\mathrm{T}} F_d - F_d = -\Pi_{r_3} F_d = \mathrm{Skew}^2(r_3) F_d = -\|F_d\| \mathrm{Skew}^2(r_3) r_{3d}$.

REMARK 2 The expression of $r_{3d} = -\frac{F_d}{\|F_d\|}$ requests $\|F_d\| \neq 0$. Due to the fact that $f = \|r_3 r_3^{\mathrm{T}} F_d\|$ represents the total thrust on the body, $\|r_3 r_3^{\mathrm{T}} F_d\|$ is generally nonzero to overcome the gravity. Therefore, $\|F_d\| \neq 0$ [10].

The Lyapunov function for the virtual system (17) is $V_3 = \frac{h_p}{2} z_p^{\mathrm{T}} z_p + \frac{h_v}{2} z_v^{\mathrm{T}} z_v + \frac{h_r}{2} z_r^{\mathrm{T}} z_r + \frac{h_{K_v}}{2} \tilde{E}_{K_v}^{\mathrm{T}} \tilde{E}_{K_v}$, where h_r is a positive constant. In order to calculate $\dot{V}_3$, $\dot{F}_d$ is needed, and its expression is given by

$$\dot{F}_d = \dot{F}_{da} - C_p d_X \tag{18}$$

where $\dot{F}_{da}$ and C_p are given by

$$\begin{aligned}
\dot{F}_{da} &= m \left(\frac{\mathrm{diag}(v) \dot{E}_{K_v}}{m} + \frac{\mathrm{diag}\left(\hat{E}_{K_v}\right) \dot{v}}{m} - \frac{h_p}{h_v} \left(v - \dot{X}_d\right)\right.\\
&\quad + \dddot{X}_d - \left(\frac{k_1}{h_p} + \frac{d^2_{X\max}}{2}\right) \left(\dot{v} - \ddot{X}_d\right) - \left(\frac{k_2}{h_v}\right.
\end{aligned}$$

$$+\frac{1}{2}\left(\frac{d_{F\max}^2}{m}+\left(\frac{k_1}{h_p}+\frac{d_{X\max}^2}{2}d_{X\max}^2\right)\right.$$

$$\left.\left.\left.\cdot\left(\dot{v}-\ddot{X}_d+\left(\frac{k_1}{h_p}+\frac{d_{X\max}^2}{2}\right)(v-\dot{X}_d)\right)\right)\right)\right)$$

and

$$C_p=m\left(\frac{h_p}{h_v}+\left(\frac{k_2}{h_v}+\frac{1}{2}\left(\frac{d_{X\max}^2}{2}+\left(\frac{k_1}{h_p}+\frac{d_{X\max}^2}{2}\right)\right.\right.\right.$$

$$\left.\left.\left.\cdot\, d_{X\max}^2\right)\left(\frac{k_1}{h_p}+\frac{d_{X\max}^2}{2}\right)\right)\right).$$

Thus, $\dot{r}_{3d}$ can be written as follows according to (18):

$$\dot{r}_{3d}=\dot{r}_{3da}+\frac{\Pi_{r_{3d}}\left(C_p d_X\right)}{\|F_d\|} \tag{19}$$

where $\dot{r}_{3d}=-\frac{\Pi_{r_{3d}}(\dot{F}_{da})}{\|F_d\|}$. Then, the derivative of V_3 is offered by the following equation according to the fact that $\dot{r}_3$ and $\dot{r}_{3d}$ are vertical to r_3 and r_{3d}, respectively,

$$\begin{aligned}
\dot{V}_3 &= \dot{V}_2+h_r z_r^{\mathrm{T}}(\dot{r}_3-\dot{r}_{3d})+\frac{h_v}{m}z_v^{\mathrm{T}}(F-F_d)\\
&= h_r(r_3-r_{3d})^{\mathrm{T}}(\dot{r}_3-\dot{r}_{3d})-\frac{h_v\|F_d\|}{m}z_v^{\mathrm{T}}\mathrm{Skew}^2(r_3)\,r_{3d}\\
&\quad+\dot{V}_2\\
&= -h_r r_3^{\mathrm{T}}\dot{r}_{3d}-h_r r_{3d}^{\mathrm{T}}\dot{r}_3-\frac{h_v\|F_d\|}{m}r_{3d}^{\mathrm{T}}\mathrm{Skew}^2(r_3)\,z_v+\dot{V}_2\\
&= -h_r r_3^{\mathrm{T}}\Pi_{r_{3d}}(\dot{r}_{3d})+h_r r_{3d}^{\mathrm{T}}R\mathrm{Skew}(e_3)\,\Omega_d-\frac{h_v\|F_d\|}{m}\\
&\quad\cdot r_{3d}^{\mathrm{T}}\mathrm{Skew}^2(r_3)\,z_v+\dot{V}_2\\
&= -h_r(\mathrm{Skew}(r_{3d})\,r_3)^{\mathrm{T}}\,\mathrm{Skew}(r_{3d})\,\dot{r}_{3d}+h_r r_{3d}^{\mathrm{T}}R\\
&\quad\cdot\mathrm{Skew}(e_3)\,\Omega_d-\frac{h_v\|F_d\|}{m}r_{3d}^{\mathrm{T}}\mathrm{Skew}^2(r_3)\,z_v+\dot{V}_2\\
&= h_r(\mathrm{Skew}(r_3)\,r_{3d})^{\mathrm{T}}\,\mathrm{Skew}(r_{3d})\,\dot{r}_{3d}+h_r r_{3d}^{\mathrm{T}}R\\
&\quad\cdot\mathrm{Skew}(e_3)\,\Omega_d-\frac{h_v\|F_d\|}{m}r_{3d}^{\mathrm{T}}\mathrm{Skew}^2(r_3)\,z_v+\dot{V}_2\\
&= -h_r r_{3d}^{\mathrm{T}}\mathrm{Skew}(r_3)\,\mathrm{Skew}(r_{3d})\,\dot{r}_{3d}+h_r r_{3d}^{\mathrm{T}}R\\
&\quad\cdot\mathrm{Skew}(e_3)\,\Omega_d-\frac{h_v\|F_d\|}{m}r_{3d}^{\mathrm{T}}\mathrm{Skew}^2(r_3)\,z_v+\dot{V}_2.
\end{aligned} \tag{20}$$

LEMMA 1 $RSkew(e_3)R^{\mathrm{T}}=Skew(r_3)$.

PROOF For any vector $q\in\Re_3$, it meets the equation that

$$\begin{aligned}
RSkew(e_3)\,R^{\mathrm{T}}q &= [r_1,r_2,r_3]Skew(e_3)\,[r_1,r_2,r_3]^{\mathrm{T}}q\\
&= \left(r_2 r_1^{\mathrm{T}}-r_1 r_2^{\mathrm{T}}\right)q.
\end{aligned}$$

Also, $r_3=r_1\times r_2$, so $Skew(r_3)q$ can be written as follows:

$$\begin{aligned}
Skew(r_3)\,q &= (r_1\times r_2)\times q\\
&= -q\times(r_1\times r_2)\\
&= -\left(r_1 r_2^{\mathrm{T}}-r_2 r_1^{\mathrm{T}}\right)q\\
&= RSkew(e_3)\,R^{\mathrm{T}}q.
\end{aligned}$$

Therefore, $RSkew(e_3)R^{\mathrm{T}}=Skew(r_3)$

According to Lemma 1 and (19), (20) can be obtained as follows:

$$\begin{aligned}
\dot{V}_3 &= \dot{V}_2+h_r r_{3d}^{\mathrm{T}}R\mathrm{Skew}(e_3)\left(\Omega_d-R^{\mathrm{T}}\mathrm{Skew}(r_{3d})\,\dot{r}_{3da}\right.\\
&\quad-R^{\mathrm{T}}\mathrm{Skew}(r_{3d})\,\Pi_{r_{3d}}\left(\frac{C_p d_X}{\|F_d\|}\right)-\frac{h_v\|F_d\|}{h_r m}\\
&\quad\left.\cdot\,\mathrm{Skew}(e_3)\,R^{\mathrm{T}}z_v\right).
\end{aligned} \tag{21}$$

The virtual control input is designed as follows:

$$\begin{aligned}
\Omega_d &= R^{\mathrm{T}}\mathrm{Skew}(r_{3d})\,\dot{r}_{3da}+\frac{h_v\|F_d\|}{h_r m}\mathrm{Skew}(e_3)\,R^{\mathrm{T}}z_v\\
&\quad+\frac{k_3}{h_r}\mathrm{Skew}(e_3)\,R^{\mathrm{T}}r_{3d}-\frac{C_p d_{X\max}^2}{2\|F_d\|}\left(r_{3d}^{\mathrm{T}}R\mathrm{Skew}(e_3)\right.\\
&\quad\left.\cdot\,R^{\mathrm{T}}\mathrm{Skew}(r_{3d})\,\Pi_{r_{3d}}\right)^{\mathrm{T}}
\end{aligned} \tag{22}$$

where the control gain k_3 is a positive constant. Then, (22) makes (21) be written as follows: $\dot{V}_3\le -k_1 z_p^{\mathrm{T}}z_p-k_2 z_v^{\mathrm{T}}z_v-k_3\|\Pi_{r_3}z_r\|^2+\frac{h_p}{2}+\frac{h_v}{2m}+\frac{h_v}{2}(\frac{k_1}{h_p}+\frac{d_{X\max}^2}{2})+\frac{h_r C_p}{2\|F_d\|}$.

REMARK 3 According to traditional backstepping methods [12], [15], the errors are defined as $z_\phi=\phi-\phi_d$ and $z_\theta=\theta-\theta_d$, where ϕ_d and θ_d are desired Eular angles. There exists the problem of singularities resulted from using inverse trigonometric functions: $\phi_d=\arcsin(P_1)$ and $\theta_d=\arctan(P_2)$, where the expressions of P_1 and P_2 can be found in [15]. Therefore, choosing r_3 to take place of Eular angles in Step 3 can avoid the abovementioned problem.

Step 4: The tracking error $z_\Omega=\Omega-\Omega_d$ is defined for this step. The virtual system is written as follows:

$$\begin{aligned}
\dot{z}_p &= v_d+d_X-\dot{X}_d+z_v\\
\dot{z}_v &= \frac{F_d}{m}-\frac{K_v v}{m}+\frac{d_F}{m}+ge_3-\ddot{X}_d+\left(\frac{k_1}{h_p}+\frac{d_{X\max}^2}{2}\right)\\
&\quad\cdot\left(v+d_X-\dot{X}_d\right)+\frac{F-F_d}{m}\\
\dot{z}_r &= -R\mathrm{Skew}(e_3)\,(\Omega_d+z_\Omega)-\dot{r}_{3d}\\
J\dot{z}_\Omega &= -\Omega\times J\Omega+\mathrm{diag}\left(l,l,\frac{1}{c}\right)M_1+d_M-J\dot{\Omega}_d
\end{aligned}$$

where the control input is designed in the form of $M_1=\mathrm{diag}(\frac{1}{l},\frac{1}{l},\hat{c})u$ and $u=[u_1,u_2,u_3]^{\mathrm{T}}$. $\hat{c}$ is the estimation of c and its error is defined as $\tilde{c}=c-\hat{c}$. The Lyapunov function for the virtual system is written as follows:

$$\begin{aligned}
V_4 &= \frac{h_p}{2}z_p^{\mathrm{T}}z_p+\frac{h_v}{2}z_v^{\mathrm{T}}z_v+\frac{h_r}{2}z_r^{\mathrm{T}}z_r+\frac{h_\Omega}{2}z_\Omega^{\mathrm{T}}Jz_\Omega\\
&\quad+\frac{h_{K_v}}{2}\tilde{E}_{K_v}^{\mathrm{T}}\tilde{E}_{K_v}+\frac{h_c\tilde{c}^2}{2c}+\frac{h_J}{2}\tilde{E}_J^{\mathrm{T}}J\tilde{E}_J
\end{aligned} \tag{23}$$

where $\hat{E}_J=[\hat{I}_X,\hat{I}_y,\hat{I}_z]^{\mathrm{T}}$ is the estimation of $J=\mathrm{diag}(E_J)=\mathrm{diag}([I_x,I_y,I_z]^{\mathrm{T}})$ and its error of estimation is $\tilde{E}_J=E_J-\hat{E}_J$. h_Ω, h_c, and h_J are positive constants. Following the calculation of $\dot{V}_3$ in the third step, $\dot{V}_4$ for this phase can be obtained as follows:

$$\begin{aligned}
\dot{V}_4 &= \dot{V}_3+h_\Omega z_\Omega^{\mathrm{T}}\left(J\dot{\Omega}-J\dot{\Omega}_d\right)+h_r r_{3d}^{\mathrm{T}}R\mathrm{Skew}(e_3)\,z_\Omega\\
&\quad-h_c\frac{\tilde{c}\dot{\hat{c}}}{c}-h_J\tilde{E}_J^{\mathrm{T}}\dot{\hat{E}}_J
\end{aligned}$$

$$= \dot{V}_3 + h_\Omega z_\Omega^{\mathrm{T}} \left(-\Omega \times J\Omega + \operatorname{diag}\left(1, 1, \frac{\hat{c}}{c}\right) u + d_M - J\dot{\Omega}_d - \frac{h_r}{h_\Omega} \operatorname{Skew}(e_3) r_{3d} \right) - \frac{h_c \tilde{c}\dot{\hat{c}}}{c} - h_J \tilde{E}_J^{\mathrm{T}} \dot{\hat{E}}_J$$
$$= \dot{V}_3 + h_\Omega z_\Omega^{\mathrm{T}} \left(-\Omega \times J\Omega + u + d_M - J\dot{\Omega}_d - \frac{h_r}{h_\Omega} \cdot \operatorname{Skew}(e_3) r_{3d} \right) - \frac{h_c \tilde{c}\dot{\hat{c}}}{c} - h_J \tilde{E}_J^{\mathrm{T}} \dot{\hat{E}}_J - h_\Omega z_\Omega(3) \frac{c\tilde{u}_3}{c} \tag{24}$$

where $z_\Omega(3)$ is the third element of z_Ω. The control input and adaptive laws are provided as follows:

$$u = \Omega \times \operatorname{diag}(\Omega) \hat{E}_J + \operatorname{diag}\left(\dot{\Omega}_d\right) \hat{E}_J + \frac{h_r}{h_\Omega} \operatorname{Skew}(e_3) \cdot R^{\mathrm{T}} r_{3d} - \frac{d_{M\max}^2}{2} z_\Omega - \frac{k_4}{h_\Omega} z_\Omega$$
$$\dot{\hat{c}} = -\frac{h_\Omega}{h_c} z_\Omega(3) u_3$$
$$\dot{\hat{E}}_J = -\frac{h_\Omega}{h_J} \left(\operatorname{Skew}(\Omega) \operatorname{diag}(\Omega) + \operatorname{diag}\left(\dot{\Omega}_d\right) \right)^{\mathrm{T}} z_\Omega. \tag{25}$$

After substituting (25) into (24), $\dot{V}_4$ can be written as follows:

$$\dot{V}_4 = \dot{V}_3 - k_4 z_\Omega^{\mathrm{T}} z_\Omega + h_\Omega z_\Omega^{\mathrm{T}} d_M - \frac{h_\Omega d_{M\max}^2 \|z_\Omega\|_2^2}{2}$$
$$\leq \dot{V}_3 - k_4 z_\Omega^{\mathrm{T}} z_\Omega + \frac{h_\Omega}{2}$$
$$\leq -k_1 z_p^{\mathrm{T}} z_p - k_2 z_v^{\mathrm{T}} z_v - k_3 \|\Pi_{r_3} z_r\|^2 - k_4 z_\Omega^{\mathrm{T}} z_\Omega + \frac{h_p}{2} + \frac{h_v}{2m} + \frac{h_v}{2}\left(\frac{k_1}{h_p} + \frac{d_{X\max}^2}{2}\right) + \frac{h_r C_p}{2\|F_d\|} + \frac{h_\Omega}{2}. \tag{26}$$

REMARK 4 The terms that contain $d_{X\max}^2 z_p$, $d_{X\max}^2 z_v$ and $d_{F\max}^2 z_v$, $d_{X\max}^2 r_{3d}$, and $d_{M\max}^2 z_\Omega$ in (10), (14), (22), and (25), respectively, take place of signum function in traditional robust adaptive control [15]. Thus, the dynamic system meets Lipschitz condition and the existence of $\dot{v}_d$, $\dot{F}_d$, and $\dot{\Omega}_d$ can be guaranteed.

To summarize, steps 1–4 are divided into two parts. The first part is steps 1 and 2. The corresponding kinetic model for the first part is (4), namely, the position model. The controller F_d are designed in the first part, so that X can track its desired trajectory X_d if the quadrotor is fully actuated. However, the direction of F_d is restricted and, thus, the attitude controller is needed in the second part. The second part is steps 3 and 4, and its corresponding kinetic model is (7). The controllers M_1 is designed in the second part, so that r_3 can track its desired trajectory r_{3d}. It is important to note that the two parts are not independent, and the desired trajectory of the second part r_{3d} is calculated by F_d in the first part. Therefore, the controller of two parts should be combined, and the final controller can be written based on steps 1–4 as follows:

$$f = \|F_d\|$$
$$M_1 = = \operatorname{diag}\left(\frac{1}{l}, \frac{1}{l}, \hat{c}\right) u \tag{27}$$

where F_d and u are given by the following equations:

$$F_d = m\left(\frac{\operatorname{diag}(v)\hat{E}_{K_v}}{m} - ge_3 - \frac{h_p}{h_v} z_p + \ddot{X}_d - \left(\frac{k_1}{h_p} + \frac{d_{X\max}^2}{2}\right)(v - \dot{X}_d) - \frac{k_2}{h_v} z_v - \frac{1}{2}\left(\frac{d_{F\max}^2}{m} + \left(\frac{k_1}{h_p} + \frac{d_{X\max}^2}{2}\right) d_{X\max}^2\right) z_v\right)$$
$$u = \Omega \times \operatorname{diag}(\Omega) \hat{E}_J + \operatorname{diag}\left(\dot{\Omega}_d\right) \hat{E}_J + \frac{h_r}{h_\Omega} \operatorname{Skew}(e_3) \cdot R^{\mathrm{T}} r_{3d} - \frac{d_{M\max}^2}{2} z_\Omega - \frac{k_4}{h_\Omega} z_\Omega. \tag{28}$$

The adaptive laws are written as follows:

$$\dot{\hat{E}}_{K_v} = -\frac{h_v}{h_{K_v} m} \operatorname{diag}(v) z_v$$
$$\dot{\hat{c}} = -\frac{h_\Omega}{h_c} z_\Omega(3) u_3$$
$$\dot{\hat{E}}_J = -\frac{h_\Omega}{h_J} \left(\operatorname{Skew}(\Omega) \operatorname{diag}(\Omega) + \operatorname{diag}\left(\dot{\Omega}_d\right) \right)^{\mathrm{T}} z_\Omega. \tag{29}$$

REMARK 5 The controller u in (28) has a term generated by backstepping technique $\dot{\Omega}_d$. The term is too complex to find the analytic form. Thus, $\dot{\Omega}_d$ can be estimated by finite difference time approximation [10]. For example, $\dot{\Omega}_d \approx \frac{\Delta\Omega_d}{\Delta t}$, where $\Delta\Omega_d$ is the change in the value of signal and Δt is the change in time.

REMARK 6 The virtual control inputs f and u can be obtained by following steps 1–4. According to (8) and (9), the actual control input F_i can be written as follows:

$$\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} f \\ M_{1x} \\ M_{1y} \\ M_{1z} \end{bmatrix} \tag{30}$$

where $M_1 = [M_{1x}, M_{1y}, M_{1z}]^{\mathrm{T}}$. In fact, it is possible to obtain $F_i < 0$ if the control gains are not chosen suitably, as shown in (30). Meanwhile, if $M_{1x,y,z}$ is relatively smaller than f, such as $(f > \sqrt{2}|M_{1x}| + \sqrt{2}|M_{1y}| + |M_{1z}|)$, F_{1-4} will be larger than 0. Therefore, $M_{1x,y,z}$ should not be too larger to cause the problem, which means the control gains $k_{3,4}$ should not be chosen too large values.

B. Closed-Loop Stability

The adaptive laws in (29) are not able to keep the signals of estimation bounded, including $\hat{K}_x$, $\hat{K}_y$, $\hat{K}_z$, $\hat{c}$, $\hat{I}_x$, $\hat{I}_y$, and $\hat{I}_z$. Therefore, the projection operators are introduced to limit the variation of estimation within the known actual range.

For any estimation signal $\hat{e} \in \{\hat{K}_x, \hat{K}_y, \hat{K}_z, \hat{c}, \hat{I}_x, \hat{I}_y, \hat{I}_z\}$, its value range is $L_e < \hat{e} < R_e$. The adaptive update law of $\hat{e}$ designed in (29) is modified by following projection algorithm:

$$\dot{\hat{e}} = \begin{cases} \dot{\hat{e}}_d, & \text{if} \quad L_e < \hat{e} < R_e \\ & \text{or} \quad \hat{e} = L_e \quad \text{and} \quad \dot{\hat{e}}_d > 0 \\ & \text{or} \quad \hat{e} = R_e \quad \text{and} \quad \dot{\hat{e}}_d < 0 \\ 0, & \text{if} \quad \hat{e} = L_e \quad \text{and} \quad \dot{\hat{e}}_d \leq 0 \\ & \text{or} \quad \hat{e} = R_e \quad \text{and} \quad \dot{\hat{e}}_d \geq 0 \end{cases} \tag{31}$$

where $\dot{\hat{e}}_d$ is the adaptive law of $\hat{e}$ in (29).

The stability of the closed-loop system under the action of projection operators is given by Theorem 1.

THEOREM 1 For any estimation signal $\hat{e} \in \{\hat{K}_x, \hat{K}_y, \hat{K}_z, \hat{c}, \hat{I}_x, \hat{I}_y, \hat{I}_z\}$, if its initial value $\hat{e}(0)$ is chosen between L_e and R_e, $\hat{e}$ will vary between L_e and R_e all the time. Furthermore, the upper bound of $\dot{V}_{1,2,3,4}$ discussed in steps 1–4 will still hold

PROOF If $\hat{e}(0)$ is chosen between L_e and R_e, then $\hat{e}$ will vary in the interval all the time [35]. All of the terms contain $\tilde{e}_i$ in $\dot{V}_{1,2,3,4}$ are $\tilde{e}\dot{\hat{e}}_d - \tilde{e}\dot{\hat{e}}$, and they can be written as follows according to (31):

$$\tilde{e}\dot{\hat{e}}_d - \tilde{e}\dot{\hat{e}} = S_e\tilde{e}\dot{\hat{e}}_d \tag{32}$$

where

$$S_e = \begin{cases} 0, & \text{if} \quad L_e < \hat{e} < R_e \\ & \text{or} \quad \hat{e} = L_e \quad \text{and} \quad \dot{\hat{e}}_d > 0 \\ & \text{or} \quad \hat{e} = R_e \quad \text{and} \quad \dot{\hat{e}}_d < 0 \\ 1, & \text{if} \quad \hat{e} = L_e \quad \text{and} \quad \dot{\hat{e}}_d \leq 0 \\ & \text{or} \quad \hat{e} = R_e \quad \text{and} \quad \dot{\hat{e}}_d \geq 0. \end{cases} \tag{33}$$

If $S_e = 0$, (32) equals zero; if $S_e = 1$, there are two possibilities. If $\hat{e} = L_e$ and $\dot{\hat{e}}_d \leq 0$, then means $\tilde{e} = e - \hat{e} = e - L_e \geq 0$ and, thus, $\tilde{e}\dot{\hat{e}}_d = \tilde{e}\dot{\hat{e}}_d - \tilde{e}\dot{\hat{e}} \leq 0$. On the other hand, if $\hat{e} = R_e$ and $\dot{\hat{e}}_d \geq 0$, then means $\tilde{e} = e - \hat{e} = e - R_e \leq 0$ and, thus, $\tilde{e}\dot{\hat{e}}_d = \tilde{e}\dot{\hat{e}}_d - \tilde{e}\dot{\hat{e}} \leq 0$. In a word, (32) is nonpositive all the time. So, the introduction of the projection operators does not affect the upper bound of $\dot{V}_{1,2,3,4}$ discussed in steps 1–4.

Based on the analysis in steps 1–4, the Lyapunov function of the complete closed-loop system (4) and (7) is chosen as $V = V_4$, which includes all the tracking errors z_p, z_v, z_r, and z_Ω and all the errors of estimation, including $\tilde{E}_{K_v}$, $\tilde{c}$, and $\tilde{E}_J$. A vector $Z = [z_p^{\mathrm{T}}, z_v^{\mathrm{T}}, z_r^{\mathrm{T}}, z_\Omega^{\mathrm{T}}, \tilde{E}_{K_v}^{\mathrm{T}}, \tilde{c}, \tilde{E}_J^{\mathrm{T}}]^{\mathrm{T}}$ is introduced. Z consists of all the errors in V. The stability of the closed-loop system is stated in Theorem 2. ■

THEOREM 2 All of the tracking errors z_p, z_v, z_r, and z_Ω and estimation errors $\tilde{E}_{K_v}$, $\tilde{c}$, and $\tilde{E}_J$ are bounded. Moreover, the ultimate boundary of Z can be modified by control gains $k_{1,2,3,4}$

PROOF According to the definition of V in (23), the lower and upper bounds of V can be given by

$$p_1\|Z\|_2^2 \leq V \leq p_2\|Z\|_2^2$$

where p_1 and p_2 are the maximum and minimum of $\{\frac{h_p}{2}, \frac{h_v}{2}, \frac{h_r}{2}, \frac{h_\Omega}{2}, \frac{h_{K_v}}{2}, \frac{h_c}{2c}, \frac{h_J I_x}{2}, \frac{h_J I_y}{2}, \frac{h_J I_z}{2}\}$. Also, the upper of $\dot{V}$ is offered in (26). Because of the fact of Theorem 1 that all the signals of estimation are bounded and the magnitude of their errors $|\tilde{e}| \leq R_e - L_e$, the upper bound of $\dot{V}$ can be written as follows:

$$\begin{aligned} \dot{V} \leq & -k_1 z_p^{\mathrm{T}} z_p - k_2 z_v^{\mathrm{T}} z_v - k_3\|\Pi_{r_3} z_r\|^2 - k_4 z_\Omega^{\mathrm{T}} z_\Omega \\ & + \frac{h_p}{2} + \frac{h_v}{2m} + \frac{h_v}{2}\left(\frac{k_1}{h_p} + \frac{d_{X\max}^2}{2}\right) + \frac{h_r C_p}{2\|F_d\|} \\ & + \frac{h_\Omega}{2} \\ \leq & -k_1 z_p^{\mathrm{T}} z_p - k_2 z_v^{\mathrm{T}} z_v - k_3 z_r^{\mathrm{T}} z_r - k_4 z_\Omega^{\mathrm{T}} z_\Omega \\ & - k_{K_v}\tilde{E}_{K_v}^{\mathrm{T}}\tilde{E}_{K_v} - k_c\tilde{c}^2 - k_J\tilde{E}_J^{\mathrm{T}}\tilde{E}_J + B_e \\ \leq & -p_3\|Z\|_2^2 + B_e \end{aligned} \tag{34}$$

where k_{K_v}, k_c, and k_J are all arbitrary positive constants. p_3 is the minimum of $\{k_1, k_2, k_3, k_4, k_{K_v}, k_c, k_J\}$. Meanwhile, the positive constant B_e is defined as follows:

$$\begin{aligned} B_e = & \frac{h_p}{2} + \frac{h_v}{2m} + \frac{h_v}{2}\left(\frac{k_1}{h_p} + \frac{d_{X\max}^2}{2}\right) + \frac{h_r C_p}{2\|F_d\|} + \frac{h_\Omega}{2} \\ & + k_4 + k_{K_v}\left(\left(R_{K_x} - L_{K_x}\right)^2 + \left(R_{K_y} - L_{K_y}\right)^2\right. \\ & \left. + \left(R_{K_z} - L_{K_z}\right)^2\right) + k_c\left(R_c - L_c\right)^2 + k_J\left(\left(R_{I_x}\right.\right. \\ & \left.\left. - L_{I_x}\right)^2 + \left(R_{I_y} - L_{I_y}\right)^2 + \left(R_{I_z} - L_{I_z}\right)^2\right). \end{aligned}$$

Therefore, $\dot{V}$ in (34) meets the condition that

$$\dot{V} \leq -\frac{p_3}{2}\|Z\|_2^2, \ \text{if } \|Z\| \geq \sqrt{\frac{2B_e}{p_3}}.$$

According to [30, Th. 4.18], $\|Z\|$ is ultimate bounded, and its ultimate boundary is given by $\sqrt{\frac{2B_e}{p_3}}$, which can be modified by p_3. ■

TABLE I
Parameters and Values of the Plant to Be Controlled

Notation	Values
m	1.465 kg
I_x	2.440×10^{-2} kg·m^2
I_y	2.400×10^{-2} kg·m^2
I_z	4.360×10^{-2} kg·m^2
l	0.160 m
$K_{x,y,z}$	0.010 for $i = 1, 2, 3$
c	4.615

IV. NUMERICAL SIMULATION

In simulation, the quadrotor UAV is supposed to track a time-varying trajectory and keep the yaw angle around zero. The control scheme is compared with common methods, including PID [26], backstepping [10], geometric controller [11], and other robust adaptive method [15]. The parameters and their values are listed in Table I. The measurement noise d_X is chosen as

$$d_X = (0.5\sin(10t), 0.5\sin(10t), 0.5\sin(10t))^{\mathrm{T}}$$

TABLE II
Value Range (L_e, R_e) and Initial Value $\hat{e}(0)$

e	L_e	R_e	$\hat{e}(0)$
$K_{x,y,z}$	-2.0×10^{-2}	2.0×10^{-2}	0
I_x	$0.2 \times I_x$	$2.0 \times I_x$	$0.5 \times I_x$
I_y	$0.2 \times I_y$	$2.0 \times I_y$	$0.5 \times I_y$
I_z	$0.2 \times I_z$	$2.0 \times I_z$	$0.5 \times I_z$
c	$0.2 \times c$	$2.0 \times c$	$0.5 \times c$

TABLE III
Control Gains of the System

Notation	Values	Notation	Values
k_1	0.1	h_p	1.0×10^1
k_2	0.5	h_v	1.0×10^{-4}
k_3	8.0	h_r	1.0×10^{-3}
k_4	8.0	h_Ω	1.0×10^2
$d_{X\max}$	$0.5\sqrt{3}$	$d_{F\max}$	$0.2\sqrt{3}$
$d_{M\max}$	$0.1\sqrt{3}$		

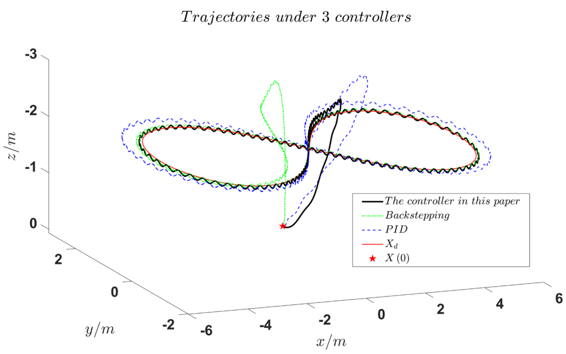

Fig. 3. 3-D trajectory tracking in simulation.

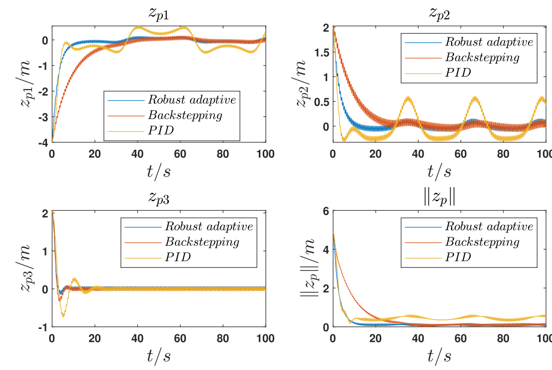

Fig. 4. Tracking errors $z_p = (z_{p1}, z_{p2}, z_{p3})^{\mathrm{T}}$ and $\|z_p\|$ in simulation.

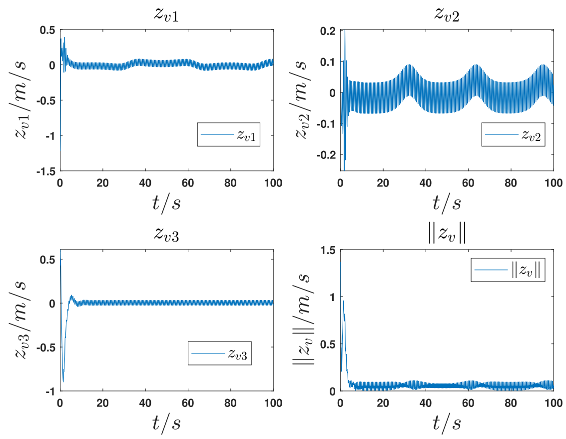

Fig. 5. Tracking errors $z_v = (z_{v1}, z_{v2}, z_{v3})^{\mathrm{T}}$ and $\|z_v\|$ in simulation.

and disturbances are

$$\begin{aligned} d_F &= (0.2\sin(10t), 0.2\sin(10t), 0.2\sin(10t))^{\mathrm{T}} \\ d_M &= (0.1\sin(10t), 0.1\sin(10t), 0.1\sin(10t))^{\mathrm{T}}. \end{aligned}$$

The value range (L_e, R_e) for any $e \in \{K_x, K_y, K_z, c, I_x, I_y, I_z\}$ is given in Table II. The control gains of the system are given in Table III. Initial conditions are assumed to be $X(0) = (1, 1, 0)^{\mathrm{T}}$, $v(0) = (0, 0, 0)^{\mathrm{T}}$, and $R(0) = \mathrm{diag}([1, 1, 1])$.

A. Comparison With PID and Backstepping Methods

In this section, the reference trajectory to be tracked is given by

$$X_d = \left(\frac{5\cos(0.1t)}{1+\sin^2(0.1t)}, \frac{5\sin(0.1t)\cos(0.1t)}{1+\sin^2(0.1t)}, -2\right)^{\mathrm{T}}. \tag{35}$$

Therefore, the desired trajectory is a Bernoulli lemniscate, which looks like ∞. There are three methods dealing with the same model and initial conditions. The simulation lasts for 100 s, and the results are shown as follows.

The desired trajectory and the trajectories of three different controllers are shown in Fig. 3 in different colors, and the controller of this article has a better performance of trajectory tracking. The details of tracking errors z_{p1}, z_{p2}, z_{p3}, and $\|z_p\|$ can be found in Fig. 4. The PID leads to $\|z_p\|$ with 0.36 m steady-state error because its feasibility is based on the linear model and is not suitable to complicated desired trajectory. Also, the transient process of the controller in this article is shorter and smoother than other two controllers'. The transient times of robust adaptive controller about four errors are 15.1, 10.5, 0.1, and 16.6 s shorter than those of backstepping method. In conclusion, compared with PID and backstepping methods, the controller of this article performs better in terms of the steady-state error and transient process. Furthermore, it is able to achieve the goal of trajectory tracking and its errors are ultimate bounded.

All other tracking errors z_v, z_r, and z_Ω are illustrated in Figs. 5–7, and they are witnessed to be bounded and vary around zero. The results in Figs. 8–10 indicate that the estimation of unknown parameters $K_{x,y,z}$, $\hat{I}_{x,y,z}$, and $\hat{c}$ are bounded. The estimation is affected by the projection operators, and all of them vary between L_e and R_e. In conclusion, all the results are consistent with Theorems 1 and 2.

Some details related to the design of controller should also be discussed. According to Fig. 11, $\|F_d\|$ varies between 14 and 20 N, and thus, it is consistent with Remark 2. The actual control inputs F_{1-4} are shown in Fig. 12. Therefore, the problem of $F_i < 0$ is successfully avoided by choosing relatively smaller control gains $k_{3,4}$ as referred in Remark 6.

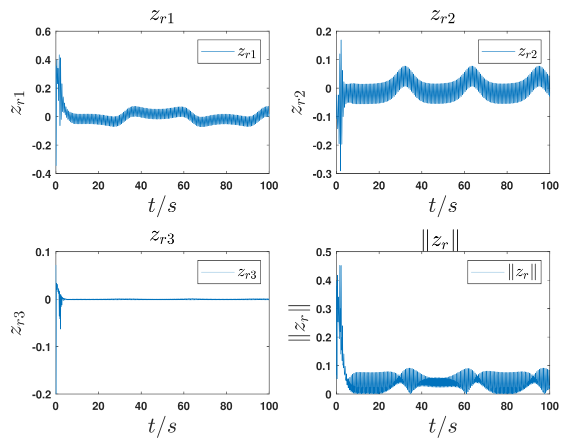

Fig. 6. Tracking errors $z_r = (z_{r1}, z_{r2}, z_{r3})^{\mathrm{T}}$ and $\|z_r\|$ in simulation.

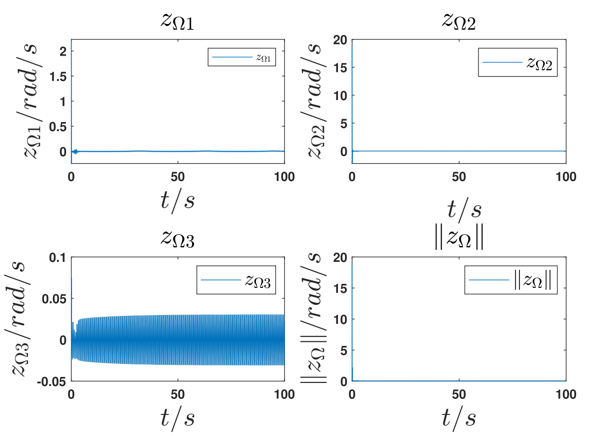

Fig. 7. Tracking errors $z_\Omega = (z_{\Omega 1}, z_{\Omega 2}, z_{\Omega 3})^{\mathrm{T}}$ and $\|z_\Omega\|$ in simulation.

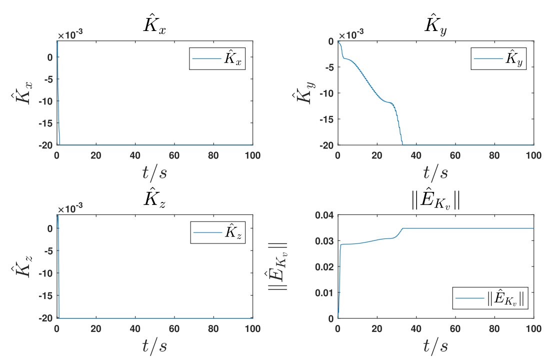

Fig. 8. Estimated parameters $\hat{E}_{K_v} = (\hat{K}_x, \hat{K}_y, \hat{K}_z)^{\mathrm{T}}$ and $\|\hat{E}_{K_v}\|$ in simulation.

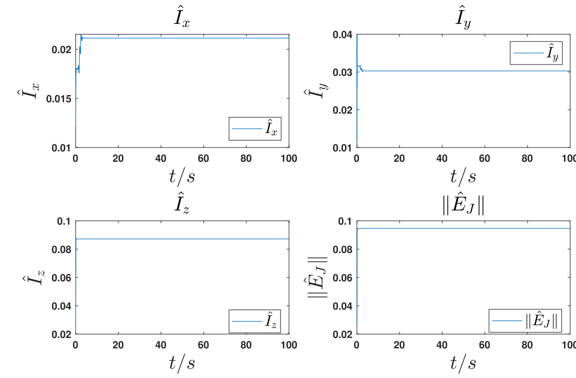

Fig. 9. Estimated parameters $\hat{E}_J = (\hat{I}_x, \hat{I}_y, \hat{I}_z)^{\mathrm{T}}$ and $\|\hat{E}_J\|$ in simulation.

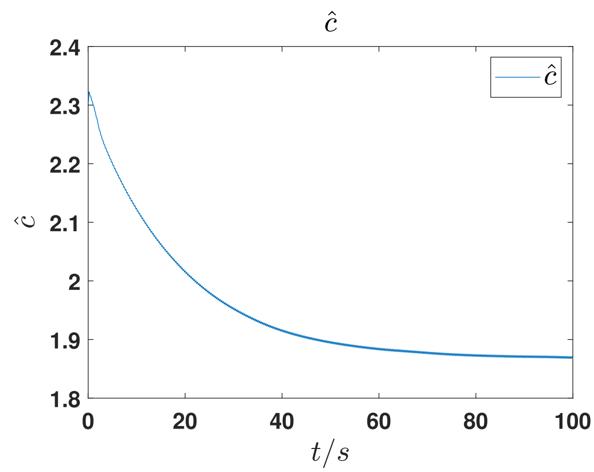

Fig. 10. Estimated parameter $\hat{c}$ in simulation.

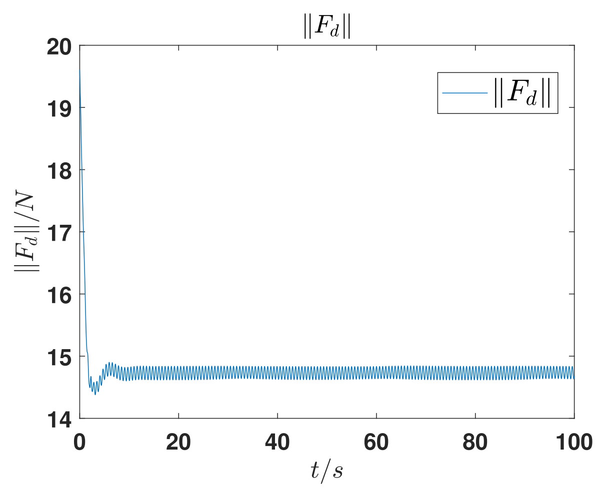

Fig. 11. Magnitude of reference resultant lift $\|F_d\|$ in simulation.

B. Comparison With Geometric Control

Geometric control is an effective way to solve trajectory-tracking problems of quadrotor UAV [11], [31]. The proposed scheme is compared with the geometric controller in [11] and their results are shown in Fig. 13. The desired trajectory and initial condition are the same with Section IV-A. Similar to the results of "PID" controller in Fig. 4, the geometric controller leads to steady-state error as well, because the position controller is designed in the form of "PID" although its attitude controller follows the way of geometric control. Furthermore, the controller in [11] fails to take disturbances, unknown parameters, and measurement noises into consideration, which accounts for worse results, including fluctuations with amplitude exceeding 0.5 in z_{p1} and z_{p2} of the geometric controller.

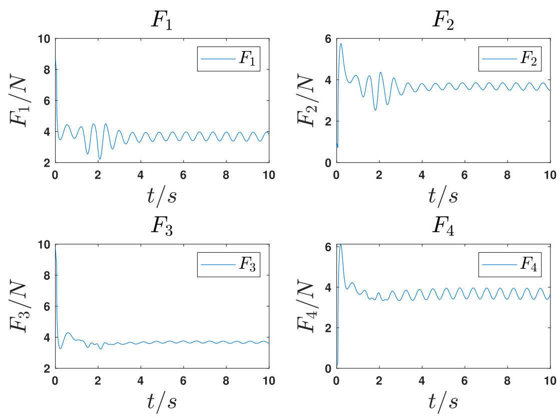

Fig. 12. Control inputs in simulation.

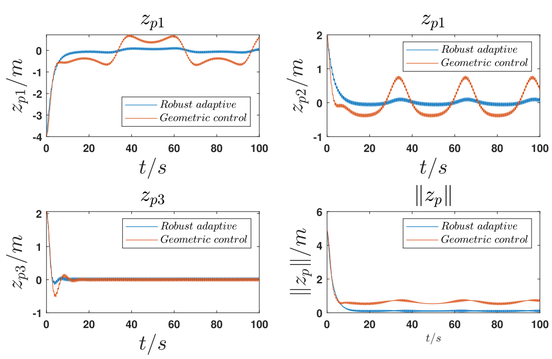

Fig. 13. Tracking errors $z_p = (z_{p1}, z_{p2}, z_{p3})^{\mathrm{T}}$ and $\|z_p\|$ in simulation: Comparison with geometric control.

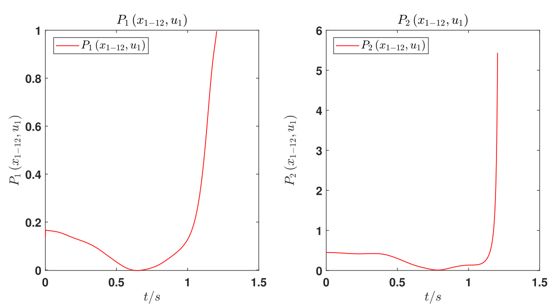

Fig. 14. Possible singularities in [15]: They can be avoided by using the controller proposed in this article.

C. Comparison With Other Robust Adaptive Method

In this section, the controller of this article is compared with other robust adaptive method in [15]. The desired trajectory and initial condition are same with Section IV-A, and the performances of two controllers are totally different. The robust adaptive controller in [15] collapse at 1.2 s, but the controller in this article successfully achieve the goal of trajectory tracking, as show in Section IV-A. According to Remark 3, the collapse is caused by the problem of singularities resulted from using inverse trigonometric functions during the calculation of desired Eular angles. The desired values of ϕ and θ are $\phi_d = \arcsin(P_1)$ and $\theta_d = \arctan(P_2)$, respectively, where the expressions of P_1 and P_2 can be found in [15]. According to Fig. 14, P_1 is over 1 after 1.2 s, which cause the singularity of ϕ_d. Compared with the failure of controller in [15], the controller in this article can avoid the problem by changing the form of errors in step 3.

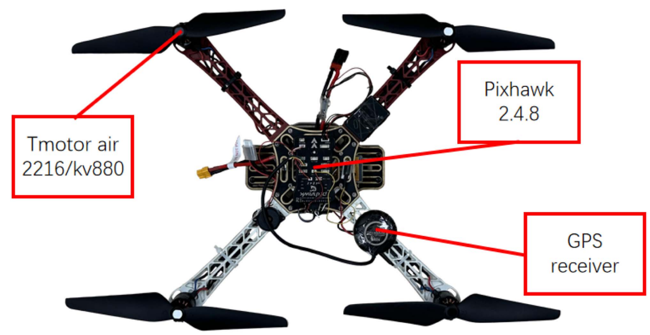

Fig. 15. Experimental quadrotor UAV testbed.

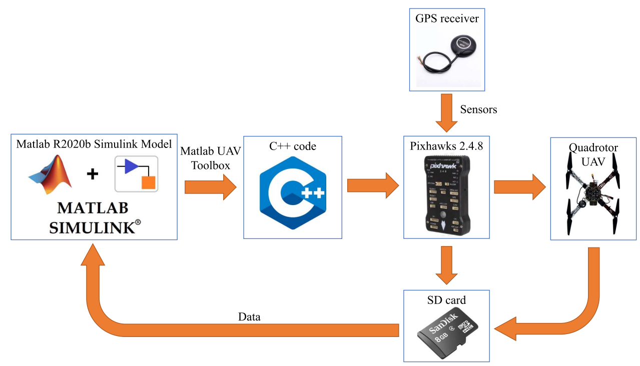

Fig. 16. Structure of the experimental testbed.

V. EXPERIMENTAL RESULTS

A. Experimental Setting

In this section, the quadrotor UAV used in the experiment is show in Fig. 15. Its framework is "DJI F450" where $l = 0.16$ m and its weighs 1.465 kg totally (including a "14.8 V, 2300 mAh" LiPo battery). Four "Tmotor Air 2216/KV880" motors work on the quadrotor. All of the signals needed by the controller except for the position information are provided by sensors integrated in a "Pixhawk 2.4.8" autopilot. Meanwhile, the position of the quadrotor in x and y-axes is offered by a GPS receiver, and its height is given by a barometer embedded in "Pixhawk 2.4.8" autopilot.

The experiment flow is shown in Fig. 16. In order to raise our working efficiency, "UAV Toolbox" of "MATLAB R2020b" is employed, and it can turn the "Simulink" model in "MATLAB" into the C++ code required by the "Pixhawk 2.4.8" autopilot directly. Then, the controller functions on the autopilot by driving four rotors. After experiments, the data are recorded in an SD card of the autopilot. Finally, the data can be read by "MATLAB."

B. Trajectory Tracking Experiment

This section presents two different trajectory tracking results under the action of the controller (27). Experiments

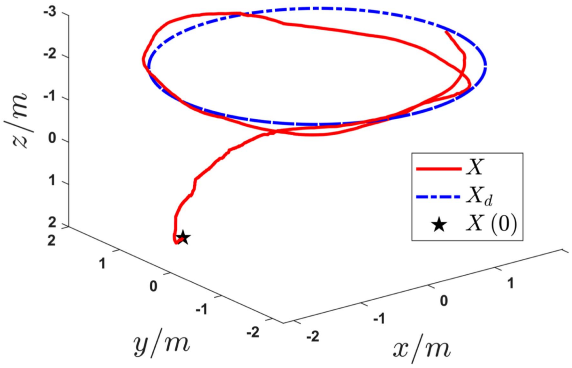

Fig. 17. Trajectory tracking of a circle curve in experiment.

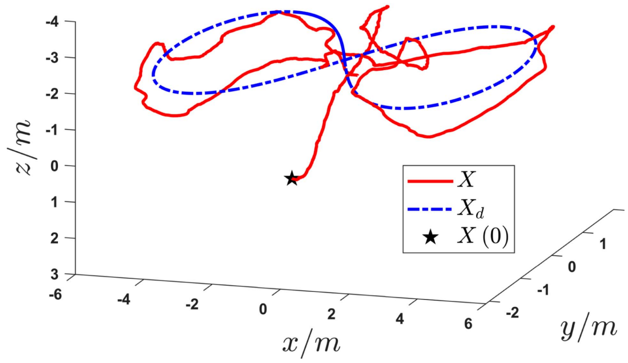

Fig. 19. Trajectory tracking of a lemniscate curve in experiment.

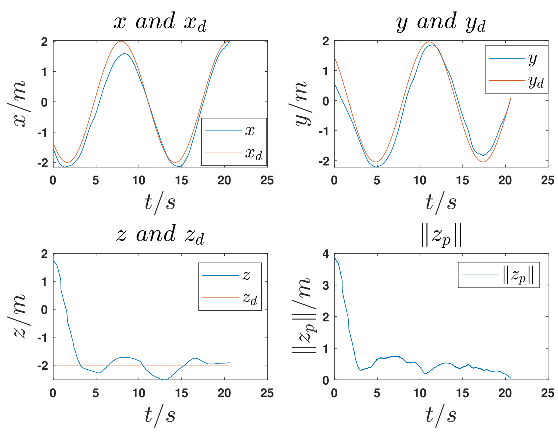

Fig. 18. X, X_d, and $\|z_p\|$ in a trajectory tracking experiment of a circle curve.

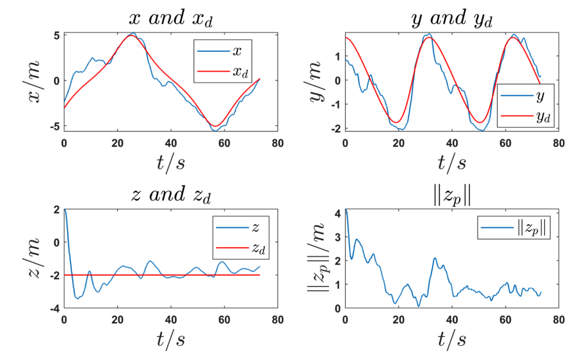

Fig. 20. X, X_d, and $\|z_p\|$ in a trajectory tracking experiment of a lemniscate curve.

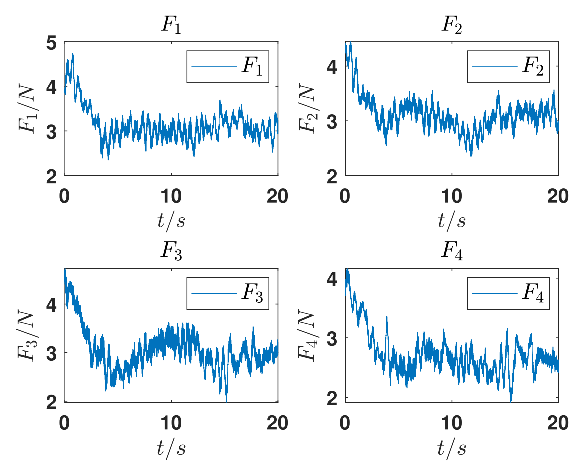

Fig. 21. Control inputs $F_{1,2,3,4}$ of tracking a circle curve in experiment.

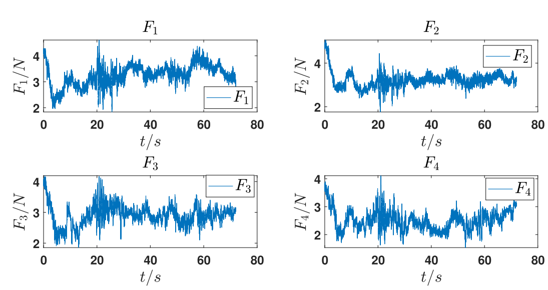

Fig. 22. Control inputs $F_{1,2,3,4}$ of tracking a lemniscate curve in experiment.

are finished outdoors, which means the quadrotor is influenced by the aerodynamic damping, disturbances, and environmental noises of measurement.

First of all, the desired trajectory is given by a circle curve as follows:

$$X_d = (2\cos(0.5t), 2\sin(0.5t), -2)^{\mathrm{T}}.$$

The related results are offered by Figs. 17 and 18. The quadrotor starts from the initial position $X(0) = (-1.6, 0.5, 1.8)^{\mathrm{T}}$ and finally converges to X_d. In addition, the quadrotor is asked to follow the lemniscate trajectory (35) and it is able to converge to X_d as well. The related results can be found in Figs. 19 and 20.

As shown in Figs. 18 and 20, the tracking error $\|z_p\|$ is ultimate bounded, which is consistent with Theorem 2. The tracking errors of $\|z_p\|$ are smaller than 1 m after 15 and 42 s, respectively. Furthermore, there are more fluctuations in z-axis than that in x and y-axes. Because, the height information is provided by the barometer, which is easier to get disturbed by noises compared with the GPS receiver as referred in Section I.

Finally, the control inputs $F_{1,2,3,4}$ of two trajectories are shown in Figs. 21 and 22. All the control inputs are smaller than 12 N, which is the upper boundary of the lift generated by the motor. Therefore, if the control gains are chosen properly, the problems of input saturation can be avoided.

VI. CONCLUSION

A robust adaptive backstepping control scheme is proposed in this article. The controller is based on a complete nonlinear kinetic model, including aerodynamic damping, disturbances, and unknown parameters that are not included in previous works. First, the proposed theoretical results rely on weaker assumptions, where the disturbances are only required to be bounded. In addition, environmental noises on the measurement of position signal is taken into consideration, so that the controller can be put into practise when doing outdoor experiments. Furthermore, the controller does not have to calculate the desired Euler angles, avoiding the singularities resulted from using inverse trigonometric functions. Finally, the proposed adaptive controller and updating laws are based on continuous functions, and thus, the closed-loop system satisfies Lipschitz condition. Theoretical proof, and simulation and experimental results show that the tracking errors are ultimate bounded, and estimation errors are bounded due to projection operators, which are consistent with theoretical results. Meanwhile, the proposed controller performs better than PID, backstepping, geometric control, and other robust adaptive method in the simulation, and its feasibility is proved in the results of experiments. Some other works inspire us to solve the problem in different ways in the future [36], [37], [38], [39], [40].

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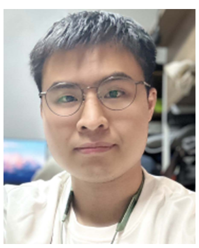

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